DSA

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For Week 4 I chose to use the Binary Search algorithm. I used two binary searches to find the last exit to the engine room based on the lower end of the range and the upper end and then filled a new array with whatever values it found. I hardcoded in the date listed in the assignment (between 10-11 on September 9th, 2234) and found one person, Violet Vale who one must assume is our killer. This took 0ms. This makes sense since it is a range of a single day and within an hour. I chose a Binary Search Tree for Week 5 and decided on DFS as my means of traversal, largely because the pseudocode for that made the most amount of immediate sense to me. This takes 468ms which also makes sense since it does not have a range like the Week 4 search and is instead traversing the entire tree in that time.

Honestly, the Binary Search Tree was the hardest part of this entire week. I'm still not entirely convinced I understand it in its entirety but I understood enough to make it work. I used a lot of visualization, favoring videos that showed the tree in a visual chart and how the search was actually working and referencing the pseudocode as I went along. I also used a method I use often when I don't understand how certain code works, which is going through and commenting out line by line and seeing what breaks (and therefore getting an idea of what it does) and trying to essentially learn it "backwards". In the end I got it to work with a fair amount of understanding. I also surprisingly had more issue with the tasks than anything else but I think that is largely because the concept of tasks was unfamiliar to me before this course so I kept forgetting how to actually make them work within Unity. This issue was largely solved by repetition.